

ADRIAN-LEONTIN RUSU

Senior Android / AAOS Engineer

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Production AAOS media & HMI • Automotive UX • Android performance & diagnostics

PROFILE

Android engineer with nearly five years of experience, currently building a production AAOS media/HMI application. **Strongest in media integration, automotive interaction, mature Android UI stacks, performance investigation and feature ownership.**

Outside production work, I build AAOS and systems projects across Binder/AIDL, JNI/Rust FFI, Media3, gRPC, PostgreSQL and modern C++ while deepening AOSP/platform knowledge.

EXPERIENCE

AscentCore | Senior Android Developer Production Android Automotive OS media / HMI

Aug 2023 - Present

- Own and deliver features in a large production AAOS media application, working across Kotlin, Java, XML, Custom Views and RecyclerView in a mature automotive codebase.
- Work across the Android media stack - MediaSession, MediaBrowser/MediaBrowserService, AudioManager/audio focus, metadata, queues, favorites/presets, source switching and transport controls - integrated with platform abstractions and proprietary AAOS system images.
- Design and debug automotive interaction for UX restrictions, parked/driving states, rotary input, steering-wheel controls, DPAD/focus navigation, hard keys, focus restoration, day/night behavior and OEM-specific RRO/resource variants.
- Investigate performance and reliability with Perfetto, ATrace, Android Studio Profiler, adb/logcat and LeakCanary: startup, main-thread blocking, frame timelines, rendering/jank, RecyclerView behavior, ANRs, Binder latency, allocations/GC and memory leaks.
- Own work through requirements, testing and integration; collaborate with QA, middleware, backend, product and architects; review PRs, support Jenkins/Gradle CI, onboarded three engineers as buddy, joined technical interviews, and use Claude/Codex for engineering and GitHub/Jira workflow automation.

Endava | Android Developer

Nov 2021 - Jun 2023

- Built Android applications in Java/Kotlin, including a POS solution integrating BLE/payment hardware, WebView flows, encrypted offline transactions and synchronization.
- Contributed to an internal Android Automotive project covering automotive UI, maps/navigation and text-to-speech integration.

Endava | Android Developer Intern

Oct 2021 - Nov 2021

CORE TOOLBOX

ANDROID & AUTOMOTIVE

Kotlin • Java • Android SDK • AAOS
Media APIs • Binder/IPC • Services • ContentProvider
UX Restrictions • Rotary/SWC/DPAD • RRO

PERFORMANCE & QUALITY

Perfetto • ATrace • Android Profiler
adb • logcat • ANR analysis • LeakCanary
JUnit • Mockito • Espresso • instrumentation • Gherkin

UI & ARCHITECTURE

XML • Custom Views • RecyclerView
RxJava • Dagger 2 • SharedPreferences • DataStore
Gradle • mature/proprietary architectures

DELIVERY & COLLABORATION

Jenkins • CI/CD • Git • PR/code review
Jira • Jama • Scrum • Kanban
QA • middleware • backend • product • architects

SELECTED ENGINEERING PROJECTS

PandaWave - AAOS Media Platform Kotlin • Compose • Media3 • AIDL • JNI • Rust

- AAOS-first media platform with modular Compose HMI, Media3/ExoPlayer, automotive restrictions, RRO-backed theming and performance infrastructure.
- Designed the Android-to-Rust boundary through Binder/AIDL, an Android service and JNI/FFI so Android owns platform execution while PandaEngine owns domain state, queues, playback intent and session coordination.

Canopy + canopy-api - Media Backend & Contract Rust • Tokio • Tonic • PostgreSQL • gRPC • Protobuf • Buf

- Rust backend for catalog/search, discovery, playback resolution, authentication, profiles, history, libraries and playlists, with PostgreSQL persistence and policy-aware repository boundaries.
- Protected playback through opaque short-lived capabilities and Nginx range streaming; versioned canopy.v1 API with Buf linting and breaking-change checks.

C++ Mastery - Modern C++ Systems Learning Lab C++26 • C17 • CMake • RAII • templates • memory management

- Hands-on C and modern C++ learning lab covering object lifetime, RAII, copy/move semantics, value categories, type deduction, templates, concepts, fold expressions and memory layout.
- Implemented an allocator-backed Vector<T> with explicit construction/destruction, alias-safe growth and exception-safety tests, plus strict warnings, AddressSanitizer and UndefinedBehaviorSanitizer.

SYSTEMS & EMERGING STACK

SYSTEMS	BACKEND & CONTRACTS	AI-ASSISTED ENGINEERING
Rust • C++ • C JNI / FFI • Tokio AOSP learning track	gRPC • Protobuf • Buf PostgreSQL • SQLx Nginx • API compatibility	Claude Code • Codex repo analysis • debugging • review custom GitHub / Jira skills

EDUCATION & DEVELOPMENT

EDUCATION	Babeş-Bolyai University, Cluj-Napoca - Computer Science
TRAINING	Claude Code 101 - completed
CURRENT FOCUS	AOSP / AAOS platform internals • modern C++ systems programming • Claude certification preparation

LANGUAGES

Romanian - Native English - Professional working proficiency